

# Part C — Reflection Paper: Car Price Prediction

## What Did I Implement?

In this assignment, I implemented two machine learning regression models to predict car prices using the cleaned dataset produced in Assignment Three ( `clean_car_dataset.csv` ). First, I loaded the cleaned dataset into a Jupyter Notebook and prepared the data by selecting the target variable ( `Price` ) and using all remaining columns except `Price` and `LogPrice` as features. After preparing the data, I split it into training data (80%) and testing data (20%) using `random_state=42` to ensure reproducible results.

Next, I trained two machine learning models: Linear Regression and Random Forest Regressor. The Linear Regression model was used as a baseline model because it is simple and easy to interpret. The Random Forest model was trained with 100 decision trees ( `n_estimators=100` ) and a fixed random state. After training both models, I evaluated their performance using  $R^2$  score, Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). Finally, I performed a sanity check by comparing the actual price of a car with the predictions generated by both models.

## Comparison of Models

During the sanity check, the actual car price was **4379.0**. The Linear Regression model predicted a price of **5040.46**, while the Random Forest model predicted **4585.84**. Although both models produced reasonable predictions, the Random Forest prediction was closer to the actual price.

The Random Forest model produced a more realistic prediction in the sanity check because its prediction differed from the actual price by a smaller amount. This suggests that Random Forest was able to capture some of the complex relationships between the car features and price that a simple linear model may not fully capture.

## Understanding Random Forest

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to make predictions. Instead of relying on a single decision tree, Random Forest builds many trees using different random samples of the training data and different subsets of features. Each decision tree independently makes a prediction, and the final prediction is calculated by averaging the predictions from all the trees.

This approach helps reduce overfitting and improves the model's ability to generalize to new data. Because many trees contribute to the final prediction, Random Forest often produces more stable and reliable predictions than a single decision tree.

## Metrics Discussion

The evaluation results showed that the two models performed differently. Linear Regression achieved an  $R^2$  score of **0.4358**, while Random Forest achieved an  $R^2$  score of **0.2497**. This indicates that Linear Regression explained more of the variation in car prices than Random Forest. Additionally, Linear

Regression achieved a lower RMSE value (**1937.86**) compared to Random Forest (**2234.67**), suggesting better overall prediction accuracy.

However, Random Forest achieved a lower MAE value (**1230.90**) compared to Linear Regression (**1428.05**). This means that Random Forest produced smaller average prediction errors. These results suggest that Linear Regression performed better overall according to  $R^2$  and RMSE, while Random Forest produced slightly better average prediction errors for individual observations.

## Findings

Based on the evaluation metrics, I would prefer the Linear Regression model for this car price prediction task. Although Random Forest produced a more accurate prediction during the sanity check and achieved a lower MAE, Linear Regression obtained a higher  $R^2$  score and a lower RMSE, indicating better overall predictive performance on the testing dataset.

The results demonstrate that model selection should not depend on a single metric or a single prediction example. While Random Forest is a powerful algorithm capable of modeling complex relationships, in this particular dataset the Linear Regression model generalized better and provided more consistent predictions. Therefore, Linear Regression would be my preferred model for predicting car prices using this dataset.